



FUEL HAND.- OPN Department
SIMHADRI



Ref: SMTTP/FUEL HAND.-/2022-23/LMI-RIM/952310

Date:

21/03/2023(16:42:13)

Subject: LMI-OPS-SYST-057_Rev0_Reliability Improvement Matrix of Coal Handling Plant

Sub: LMI - 2023 “RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT”, prepared w.r.t CC-OS issued OGN

CC-OS has issued the OGN COS-ISO-00-OGN/OPS/SYST/057, Rev. No.: 0 Jan 2023 “RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT” (copy attached for ready reference).

The same has been reviewed and in line with the OGN, **LMI, Ref NO ” LMI/OGN/OPS/SYST/057, Rev.00 “RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT”** has been prepared and is being put up for review and kind approval please.

Submitted by:

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Attached proposed LMI (LMI-OPS-SYST-057_Rev0_Reliability Improvement Matrix of Coal Handling Plant.pdf) has been reviewed and found in line with OGN: COS-ISO-00-OGN/OPS/SYST/057. submitted for kind approval please.

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Reviewed with reference to OGN: COS-ISO-00-OGN/OPS/SYST/057, found in order. Submitted for approval please.

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On:30/03/2023(19:57:43) **Ref:**SMTTP/FUEL HAND.-/2022-23/LMI-RIM/952310

Location Management Instructions

LMI/OGN/OPS/ SYST/057

Issue No.1, Revision No.0, Date: Mar'2023

(Ref: COS-ISO-00-OGN/OPS/ SYST/057 Rev. No.: 0 Jan 2023)

LMI RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT

Approved for
Implementation

.....
BUH, Simhadri

Date:

NTPC LIMITED
SIMHADRI SUPER THERMAL POWER STATION
COAL HANDLING PLANT – FUEL MANAGEMENT DEPARTMENT

LMI: RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT			
<i>LMI NO : LMI/OGN/OPS/ SYST/057</i>			
Issue No.	1	Revision : 00	Date:

RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT

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1.0 SUMMARY

1.1. Superseded Document	OIN/OPS/SYST/075, Rev. 0 Oct. 2020	
1.2. Scope	Reliability Improvement Matrix is a basis for evaluation of performance of CHP.	
1.3. Review	Review Frequency	2 yearly basis, or as and when required, whichever is lower

2.0 INTRODUCTION

Coal Handling Plant plays an important role in steady generation of power, hence its reliability is very important.

CHP Reliability Improvement Matrix (RIM) helps in improving the reliability of CHP equipment and systems by monitoring critical performance parameters. The compliance of this matrix is used as basis for evaluation of performance of CHP of all stations. Accordingly, weightages have been allocated to respective parameter for evaluation. It also enables internal benchmarking among CHP of all stations.

In order to incorporate changes arises with due course of time, monitoring of more performance parameters which is required for efficient and reliable operation & maintenance, as well as for post failure analysis a new version of RIM is introduced.

3.0 PROCESS

3.1 Frequency of Capturing RIM Data

RIM marking to be done by FM department monthly, which should be reviewed by Head of O&M. Data to be recorded every month and to be endorsed by Regional OS during ORT visit. Data recording / entering is done either on Online platform or manually through excel sheet and should be sent to COS monthly by 7th of each month, by FM Department.

3.2 Utilization of RIM Marks

RIM marks is used as PEM marks in totality in the scale of 3, for the CHP.

4.0 RECORDS TO BE MAINTAINED

RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT

Station:

Region:

Year:

Month:

Criteria	Sl. No	Item Description	Freq uency	Reference Documents	Evaluation Criteria	Weight age	Rating
A	Safety	Safety		Safety Report	If any reportable incident in month then entire marks on safety is Zero, else rating on different parameters below as A1 to A8		
		1	Deviation in PTW System	Mont hly	Safety Report	If No Deviation then full marks else 0	1.5
		2	% closure of Safety Deviation	Mont hly	SAP Report	100% closure then full marks, 99-90% closure then 1 marks, 89-70% closure then 0.5 marks, else 0	1.5
		3	Compliance to Safety Audit observations	Half Yearly	SAP Report	100% compliance then full marks, 99-90% compliance then 1 marks, 70% compliance then 0.5 marks, else 0	1.5
		4	Use of LOTO Kit for Mechanical Isolation.		Department Report	If 100% complied then full marks else 0	1.0
		5	JSA linked with PTW		PTW with JSA	If 100% complied then full marks else 0	1.0
		6	Load Testing of Lifting Tools & Tackles, Hoists & Lifts	Yearly	Department Report/Safet y Report	If 100% complied then full marks else 0	1.0
		7	Availability & Use of Safety Net along with Safety Kit (e.g. full body harness, life line, rope grab, fall arrestor) for working atheight.		Department Report	If 100% complied then full marks else 0	1.0

RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT

Criteria		Sl. No	Item Description	Frequency	Reference Documents	Evaluation Criteria	Weightage	Rating
		8	Fire Safety					
		i)	Maintaining specified Hydrant pressure at Highest & Farthest Point	Daily	Logbook/Protocol	If 100% complied then full marks else 0	0.5	
		ii)	Monthly joint inspection of FDPS System.	Monthly	Logbook/Protocol	If 100% complied then full marks else 0	0.5	
		iii)	Availability of Fire Detection system at S/R, Control Room & MCCs.		Logbook/Protocol	If 100% complied then full marks else 0	0.5	
		iv)	Availability of Fire Fighting system at Yard.		Logbook/Protocol	If 100% complied then full marks else 0	0.5	
B	Performance	1	Generation Loss (PL+FO) due to CHP Problem (SAP Code 0402)		SAP Report	If no Loss full marks, If Not Zero but $\leq 0.1\%$ the 3 marks, else 0	5.0	
		2	Generation Loss(PL+FO) due to Wet Coal (SAP Code 0413)		SAP Report	If no Loss full marks, If Not Zero but $\leq 0.1\%$ the 3 marks, else 0	5.0	
		3	Delay in rake unloading due to CHP (% of Total Delay)		FT Report / Department Report	If no delay full marks, If Not zero but less than 10% then 3 marks, else 0	5.0	
		4	Belt Utilization Factor (Monthly Average for unloading streams)		Department Report	If $\geq 70\%$ full marks, if 50-69% then 1 mark, else 0	2.0	
C	Equipment Availability	1	WT, SAC & Apron Feeder Availability		Log book/ Deptt report	If any WT/SAC/AF at station not available due to breakdown for >2 days then 0, else full marks	1.0	
		2	Stacker-Reclaimer Availability		Log book/ Deptt report	If any S/R at station not available due to breakdown for >2 days then 0, else full marks	1.0	

RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT

Criteria		Sl. No		Item Description	Freq uency	Reference Documents	Evaluation Criteria	Weight age	Rating
		3		Crusher Availability		Log book/ Deptt report	If any Crusher at station not available due to breakdown for >2 days then 0, else full marks	1.0	
		4		VSF/VGF Availability		Log book/ Deptt report	If any VSF/VGF at station not available due to breakdown for >2 days then 0, else full marks	1.0	
		5		Tripper/Bunker Conveyor Availability		Log book/ Deptt report	If any Tripper or Bunker Conveyor at station not available due to breakdown for >8 Hrs then 0, else full marks	1.0	
		6		Paddle Feeder Availability		Log book/ Deptt report	If any Paddle Feeder at station not available due to breakdown for >2 Days then 0, else full marks	1.0	
		7		Conveyor Availability		Log book/ Deptt report	If any Conveyor at station not available due to breakdown for >7 Days then 0, else full marks	1.0	
D	Practices for System Healthiness	1	i)	CHP Capacity Test	As per schedule defined in OGN/OPS/PERF/009	EEMG/ROS/ORT Report	If Not Due or conducted in current month then full marks, else 0	3.0	
			ii)	Capacity achieved in last test		EEMG/ROS/ORT Report	If capacity achieved is >=85% then full marks, If 70-84% then 1 mark, else 0	2.0	
		2	i)	Reclaiming Capability Check Reclaiming for eight hours in a	Monthly	ORT Report	If Complied then full marks, else 0	3.0	

RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT

Criteria	Sl. No	Item Description	Freq uency	Reference Documents	Evaluation Criteria	Weight age	Rating
		day as per opportunity (Month for which filled)					
		ii) Reclaiming capacity achieved		ORT Report	If capacity achieved is $\geq 90\%$ then full marks, else 0	2.0	
	3	Reliability Improvement of Wagon Tippler					
		i) Health Assessment(Equipment & Structure) of WT by OEM/OES/Expert services	Yearly	Service report	If report available then full marks, else 0	1.5	
		ii) Hydraulic system	As per schedule defined in OGN/OPS/SYST/013	Service report	If report available then full marks, else 0	1.0	
		iii) Protection Interlock Checking	Monthly	Protocol	If report available then full marks, else 0	1.0	
		iv) Availability of online oil purifier for hydraulic powerpacks of WT and SAC		Department Report	If available full marks, else 0	1.0	
		v) Measurement of NAS (Value must be maintained 7 or below).	Monthly	Department Report	If report available then full marks, else 0	1.0	
	4	Reliability Improvement of Stacker Reclaimer					
		i) Health Assessment (Equipment and	Yearly	Service Report	If report available then full marks, else 0	1.5	

RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT

Criteria	Sl. No	Item Description	Freq uency	Reference Documents	Evaluation Criteria	Weight age	Rating
		Structures) by OEM/OES/Exper t services					
	ii)	Hydraulic System Healthiness Checking	As per sched ule define d in OGN/ OPS/S YST/0 13	Service Report	If report available then full marks, else 0	0.5	
	iii)	Centralized Auto Lubrication System checking	As per sched ule define d in OGN/ OPS/S YST/0 13	Service Report	If report available then full marks, else 0	0.5	
	iv)	Slew bearing healthiness checking	Half Yearly	Protocol	If report available then full marks, else 0	0.5	
	v)	Protection/Interl ock checking	Mont hly	Protocol/Dep tt report	If report available then full marks, else 0	0.5	
	vi)	Rail clamp/Thruster brake checking	Mont hly	Department Report	If report available then full marks, else 0	0.5	
	vii)	Availability of online oil purifier for hydraulic powerpacks of S/R (Yes/No)		Chemistry Report/Lab report	If available then full marks, else 0	0.5	
	viii)	Measurement of NAS (Value must be maintained 7 or below)	Mont hly	Logbook/De partment Report	If report available then full marks, else 0	0.5	
	ix)	Slew Drive Motor Current Reading Checking	Mont hly	Logbook/De partment Report	If report available then full marks, else 0	0.5	
	x)	S/R earthing checking	Mont hly	Logbook/De partment Report	If report available then full marks, else 0	0.5	

RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT

Criteria	Sl. No	Item Description	Freq uency	Reference Documents	Evaluation Criteria	Weigh tage	Rating
	5	Crusher Reliability Improvement:					
		i) Crusher Spring Deck checking by OEM/Expert	Yearly	Service report	If report available then full marks, else 0	0.5	
		ii) Vibration Isolation System(Crusher Spring Deck) checking by department	Quart erly	Department report	If report available then full marks, else 0	0.5	
		iii) Annual Overhauling of Crusher	Yearly	Department Report	If report available then full marks, else 0	0.5	
		iv) Availability of hammers stock as per guideline		Deptt report	If available as per minimum critical spare list the full marks, else 0	0.5	
		v) Availability of online VMS (Y/N)		Deptt report	If available full marks, else 0	0.5	
		vi) Availability of Current of crusher/VSF/BF in DCS/PLC		Protocol/ Deptt report	If available full marks, else 0	0.5	
	6	Conveyor System Reliability:					
		i) Belt Mapping	Mont hly	Prescribed COS Format	If report available full marks, else 0	1.0	
		ii) Idler Mapping	Mont hly	Prescribed COS Format	If report available full marks, else 0	1.0	
		iii) Pulley Mapping	Mont hly	Prescribed COS Format	If report available full marks, else 0	1.0	
		iv) Pulley Bearing Greasing	Mont hly	Department Report	If report available full marks, else 0	1.0	
		v) Protection/Interlock checking	As per sched ule define d in OGN/ OPS/S YST/0 13	Protocol	If report available full marks, else 0	1.0	

RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT

Criteria	Sl. No	Item Description	Freq uency	Reference Documents	Evaluation Criteria	Weight age	Rating
		vi) Current and SOE in DCS/PLC		Protocol	If available full marks, else 0	1.0	
	7	Paddle Feeder Reliability Improvement					
		i) Availability of online oil purifier for hydraulic power packs of P/F		Department Report	If available full marks, else 0	1.0	
		ii) Measurement of NAS (Value must be maintained 7 or below)	Mont hly	Chemistry Report/Lab	If report available full marks, else 0	1.0	
	8	Travelling Tripper Reliability:					
		i) Availability of Over Speed Protection	Mont hly	Department Report	If available full marks, else 0	1.0	
		ii) Rail Clamp /Thruster Brake checking	Mont hly	Department Report	If report available full marks, else 0	1.0	
	9	PM Compliance of CHP Equipment	Mont hly	SAP Report/MTP Report	If 100% compliance then full marks, 80% or more then 1.5 marks, Less than 80% then 0	2.5	
	10	Compliance of Technical Audit Report	Yearly	MTP/Depart ment Report	If 100% compliance then full marks, 80% or more then 1.5 marks, Less than 80% then 0	2.5	
	11	MD, SM, MS Availability					
		i) MD in service with sensitivity level as specified	Daily	Log book/ Deptt report	If 100% available then Full Marks, else 0	0.5	
		ii) SM in service with sensitivity level as specified	Daily	Log book/ Deptt report	If 100% available then Full Marks, else 0	0.5	
		iii) ILMS/CBMS in service with sensitivity level as specified	Daily	Log book/ Deptt report	If 100% available then Full Marks, else 0	0.5	
	12	Stackyard Management					

RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT

Criteria		Sl. No		Item Description	Freq uency	Reference Documents	Evaluation Criteria	Weight age	Rating
			i)	Practice of FIFO	Mont hly	Department Report	If report available thenfull marks else 0	0.5	
			ii)	GCV mapping	Mont hly	Chemistry Report	If report available thenfull marks else 0	0.5	
			iii)	Age Analysis	Mont hly	Department Report	If report available thenfull marks else 0	0.5	
			iv)	Hot Spot Detection by Thermography	Mont hly	Department Report	If report available thenfull marks else 0	0.5	
			v)	Compaction Index	Mont hly	Department Report	If report available thenfull marks else 0	0.5	
		13		Cleanliness, House Keeping, Environment & Safe Working Conditions: Chute Leakages, Coal Spillage from Skirts/Scrappers /Sway etc. Coal Dust & Slurry Accumulation, Dust Emission, Oil Spillages, Scrap, Coal/ Oil Accumulation in Drains & Pits, Lighting/Illumina tion condition, General Cleanliness, Aesthetic Look & 5S, Safe working conditions including hand rail, toe guards, sealing of floor openings etc., Effective Operation of Coal Settling Pond with recycling of water.	Mont hly	Joint Surveillance check of mentioned points with ROS during Pre ORT visit in following locations (Each Area ofany stage sel ected Randomly):- 1. Unloading Area(TH/WT) 2. Crusher House 3. High Rise TP of Main PlantArea. 4. Coal Settling Pond. Ratingto be done as Good, Satisfactory and Poor	If overall rating by ROSis Good then 5, If Satisfactory then 3,if Poor then 0.	5.0	

RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT

Criteria		Sl. No	Item Description	Frequency	Reference Documents	Evaluation Criteria	Weightage	Rating
		14	Structure Checking & Painting			If any structure failure in current financial year then entire marks on Structure Checking & Painting would be Zero, else rating on different parameters as below D14-i) & ii)		
		i)	Structures Inspection	As per Station LMI	Department Report / Committee Report / Inspection Report	If Action Taken based on plant level committee inspection report then full marks, if inspection done by plant level committee then 2.0, else 0	2.5	
		ii)	Painting	As per Station LMI	Department Report / Committee Report / Inspection Report	If Action Taken based on plant level committee report then full marks, if inspection done by plant level committee then 2.0, else 0	2.5	
E	Rationalization of Contracts	1	Contract rationalization activity undertaken		Department Report	If contract rationalization activity undertaken then full marks, else 0	2.0	
F	Availability of critical spares	1	Availability of Conveyor Belt Stock		Department Report	100% safety stock then 2.5, 80% safety stock then 1.5, less than 80% stock then 0	2.5	
		2	Availability of recommended spares		Department Report	100% items availability then 2.5, 80% items available then 1.5, less than 80% items available then 0	2.5	
G	Monsoon Preparedness	1	Award of Monsoon Contract		Department Report	If PO by April / PR by Oct then full marks, else 0	1.0	
		2	Availability of LDPE Sheet for 100% stock pile covering.		Department Report	If available for 100% stockpile coverage by April / PO by Jan / PR by Oct then full marks, else 0	1.0	
		3	Availability of Diesel Driven Dewatering		Department Report	If available then full marks, else 0	1.0	

RELIABILITY IMPROVEMENT MATRIX OF COAL HANDLING PLANT

Criteria		Sl. No		Item Description	Frequency	Reference Documents	Evaluation Criteria	Weightage	Rating
				Pumps, Dewatering/Submersible pumps in underground locations/MCCs.					
		4		Availability of Cladding sheets & Roof Sheets in Conveyor Galleries & TPs		Department Report	If 100% available then full marks, if 90% then 0.5 marks, else 0	1.0	
H	New Technology/ New Initiatives	1		Any other innovative measures, implementation of PC/QC recommendations etc.		Department Report	If implemented then full marks, else 0	1.0	
TOTAL								100.0	0.0
Total Rating % (Evaluated Out of 100)									0